Stage 4 — Questions and Answers

Q1. Reading a hazard ratio

The Cox hazard-ratio table gives **TP53 multi-hit HR = 3.210** and **log_PLT HR = 0.832 per standard deviation**. Higher platelets are protective, while TP53 multi-hit is associated with higher hazard, all else equal.

Step 1: Convert HRs to coefficients using beta = ln(HR).

beta(TP53) = ln(3.210) = 1.166

beta(PLT) = ln(0.832) = -0.184

Step 2: Combine the coefficients. TP53 multi-hit is present (x = 1), and platelets are one SD below average (x = -1).

eta = 1.166(1) + (-0.184)(-1) = 1.350

Combined HR = exp(1.350) = 3.86

Step 3: Check by direct multiplication.

HR = 3.210 x (1 / 0.832) = **3.86**

The patient has approximately **3.86 times the instantaneous hazard of death** relative to an otherwise identical patient without TP53 multi-hit and with average platelets. This is a hazard ratio, not a probability ratio.

Q2. Why did SF3B1 stop being protective?

Stage 2 Kaplan–Meier analysis found median survival of **4.56 years** for SF3B1-mutated patients versus **2.57 years** for patients without the mutation. In the adjusted Cox model, however, SF3B1 has HR **1.02 (95% CI 0.88–1.18)**.

The explanation is **confounding**: SF3B1-mutated patients tend to have low blast percentages and favourable karyotypes, which themselves predict better survival. Stage 2 showed that blasts below 5% had the longest median survival (4.47 years), while Stage 3 showed very good cytogenetics had the longest median survival (4.44 years). Once these and other variables are held constant, SF3B1 shows little independent association; its confidence interval includes 1, consistent with no adjusted effect.

Q3. Is the elastic-net worth its complexity?

The paired centre-grouped cross-validation results show:

Comparison	Mean difference	SD	Folds won
C - B	+0.0278	0.0144	14/15
D - C	+0.0059	0.0086	10/15

Model D (elastic-net) improves on model C by only **0.0059**, smaller than the paired-difference SD of **0.0086**, and wins 10 of 15 folds. In contrast, model C improves over benchmark B by 0.0278 and wins 14 of 15 folds. Thus, the extra gain from D is modest and uncertain.

A reason to choose **D** is its slightly higher mean C-index (0.722 versus 0.716), using penalization to draw on a broader feature set. A reason to choose **C** is its 24 literature-selected, clinically interpretable variables, versus 44 non-zero coefficients from 84 candidates in D; C may also be less vulnerable to centre-specific measurement differences at KYW.

My choice is model C for a doctor-facing model: it is easier to explain and nearly as accurate. I would favour D only if its small gain persisted in independent external validation.